

Requirements Specification and Workload Model

Music Streaming Platform Inspired by Spotify

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1 Executive Summary

This document presents the initial requirements specification and workload model for a large-scale music streaming platform inspired by Spotify. The objective of this first delivery is to describe what the system must do and under which operational conditions it must operate, without selecting cloud providers, specific cloud services, databases, or implementation technologies.

The system studied covers the central capabilities of a large-scale music streaming platform: catalog ingestion and management, content availability according to rights and territories, user and subscription management, music playback, personal libraries and playlists, music recommendation, advertising for users of the free tier, listening telemetry, and derived experiences such as Wrapped. Royalty calculation, settlement, and payment are explicitly outside the scope of this project.

Publicly available information from Spotify is used whenever possible to establish realistic scale and workload references. Values that cannot be directly obtained from public Spotify sources are explicitly identified as derived calculations, lower-bound references, or proposed requirements.

2 System Description

2.1 Business Problem

The platform must allow users to discover and play music immediately while managing a global catalog whose content may be subject to restrictions related to territory, availability period, and rights. At the same time, the system must differentiate the capabilities available according to the user's plan, record listening activity, and support derived processes such as personalization, advertising, and annual listening summaries. Playback events are retained for product functionality and analytics, but royalty calculation and settlement are outside the project scope.

Spotify connects content providers and rights holders, the digital platform, and the users who consume the content. Music can reach the platform through labels, distributors, or other authorized content providers that deliver recordings, releases, metadata, and availability information [1, 2].

The main operational challenge is scale. Spotify currently reports hundreds of millions of monthly active users and operates across 184 markets [19, 21]. Its engineering organization also reports approximately 100 million concurrent clients and between 11 and 12 million backend requests per second at any given moment [16].

For this project, the system boundary is intentionally limited to the **music-streaming domain**. Podcast and audiobook catalog management are not included in the modeled system.

2.2 Actors

Actor	Type	Role	Interaction Frequency	System Expectations
Free User	Human	Registered user with access to the free tier. The user can play music according to the applicable rules and receives advertising during music playback sessions.	Frequent / daily	Fast playback, catalog availability, consistent playlists and library data, and reliable ad-supported access.
Premium User	Human	Registered user with an active Premium subscription that enables additional capabilities, including authorized offline music playback.	Frequent / daily	Low-latency playback, subscription entitlement consistency, offline access, and reliable synchronization of personal content.
Distributor / Label / Content Provider	External system / organization	Delivers music releases, recordings, metadata, artwork, and updates associated with rights, territories, and availability [1, 2].	Continuous / batch catalog updates	Reliable ingestion, validation, timely publication, and correct application of metadata and territorial availability.
Payment Provider	External system	Processes payment operations associated with subscriptions and returns the result of those operations to the platform.	On subscription and renewal events	Reliable exchange of payment results and correct subscription-state updates.
Advertising System / Advertiser	External system / organization	Participates in the advertising ecosystem used to monetize playback sessions for users of the free tier [8, 9].	Continuous during free-tier usage	Correct advertisement eligibility, timely decisions, and reliable delivery without unnecessarily interrupting music playback.

2.3 Main Use Cases

The following use cases describe the principal interactions supported by the modeled music-streaming platform. They are organized according to the functional areas represented in the use-case diagrams. Each specification identifies the participating actors, preconditions, main flow, alternative flows, and resulting postconditions.

2.3.1 Account Management

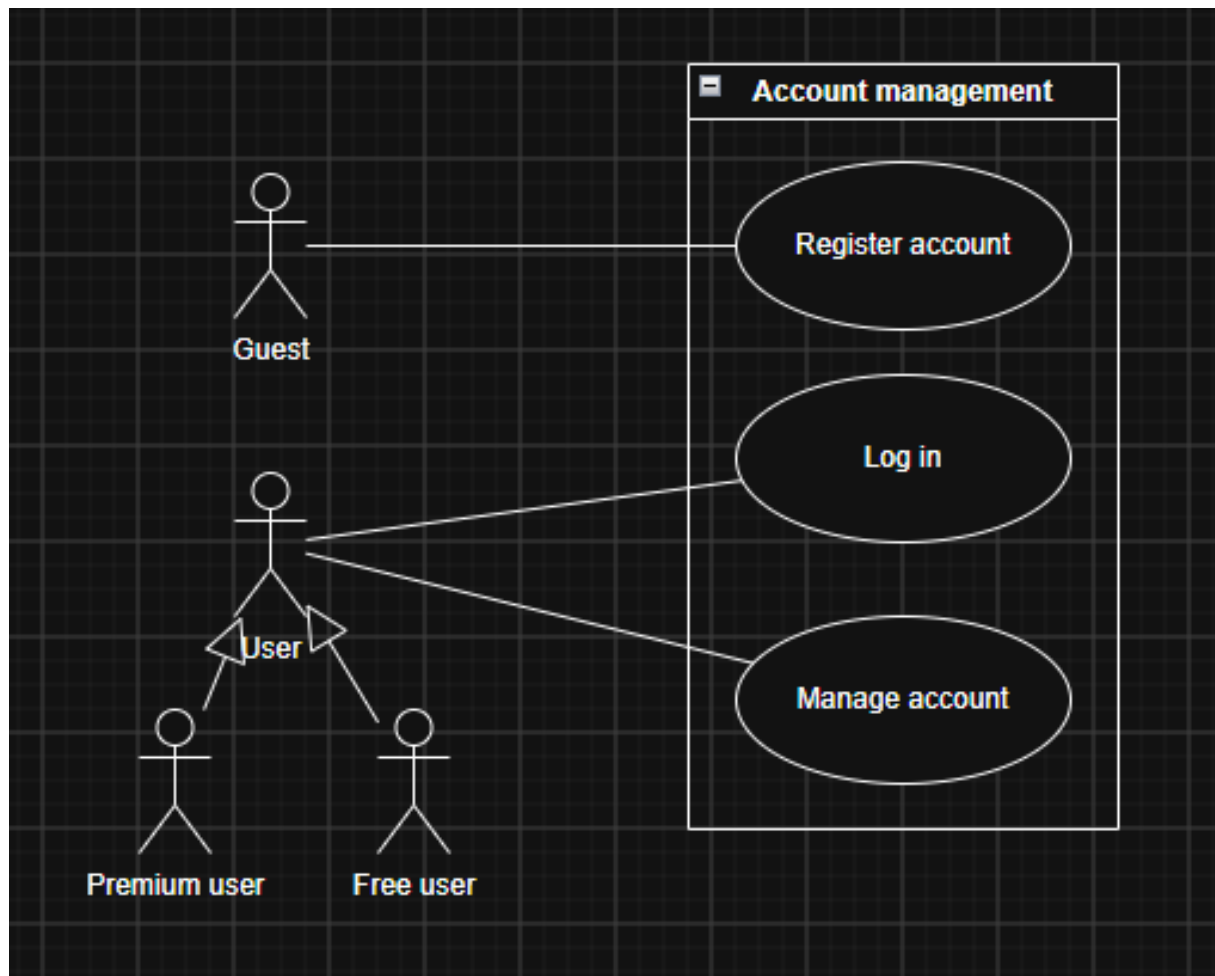


Figure 1: Use case diagram: Account Management

Use Case:	UC-AC-01 Register Account
Actors:	Guest
Preconditions:	The person does not have an authenticated session and has access to the registration functionality. The registration information required by the platform can be provided.
Main Flow:	1. The guest selects the option to create a new account. 2. The system requests the information required to register the user. 3. The guest enters the requested account information. 4. The system validates the submitted data and verifies that the account does not conflict with an existing account. 5. The system creates the new user account. 6. The system confirms that the registration was completed successfully.
Alternative Flows:	A1. If required information is missing or invalid, the system rejects the registration and requests correction of the affected fields. A2. If the supplied account identifier is already associated with an existing account, the system informs the guest and does not create a duplicate account. A3. If the registration service is temporarily unavailable, the account is not created and the guest may retry later.
Postconditions:	A new user account exists in the platform when registration succeeds. Otherwise, no new account is created.
Description:	Allows a guest to create the account required to access personalized functionality of the music-streaming platform.

Table 2: Use case specification UC-AC-01: Register Account

Use Case:	UC-AC-02 Log In
Actors:	User
Preconditions:	The user has a previously registered account and is not currently authenticated in the application.
Main Flow:	1. The user selects the login functionality. 2. The system requests the user's authentication credentials. 3. The user submits the credentials. 4. The system validates the submitted credentials. 5. The system identifies the account and its current subscription state. 6. The system establishes an authenticated session and grants access to the functionality available to the user.
Alternative Flows:	A1. If the credentials are incorrect, authentication is rejected and the user is informed that the supplied information is invalid. A2. If the account cannot currently be accessed, the system does not establish the session. A3. If authentication services are temporarily unavailable, the user is informed and may retry later.
Postconditions:	An authenticated session associated with the user's account exists when login succeeds. Otherwise, the user remains unauthenticated.
Description:	Allows an existing user to authenticate and access the functionality associated with their account and subscription type.

Table 3: Use case specification UC-AC-02: Log In

Use Case:	UC-AC-03 Manage Account
Actors:	User
Preconditions:	The user has a registered account and has successfully authenticated in the platform.
Main Flow:	1. The user opens the account-management functionality. 2. The system retrieves the current account information. 3. The user selects the information or account configuration that they want to modify. 4. The system validates the requested modification. 5. The system stores the accepted changes. 6. The system displays the updated account information.
Alternative Flows:	A1. If the supplied information is invalid, the system rejects the change and indicates which information must be corrected. A2. If a sensitive operation requires additional authentication, the system requests confirmation before applying the modification. A3. If the update cannot be completed, the previous account data remains unchanged.
Postconditions:	The account contains the confirmed changes when the operation succeeds. Unconfirmed or invalid modifications are not stored.
Description:	Allows an authenticated user to view and modify the account information and settings managed by the platform.

Table 4: Use case specification UC-AC-03: Manage Account

2.3.2 Music Consumption Management

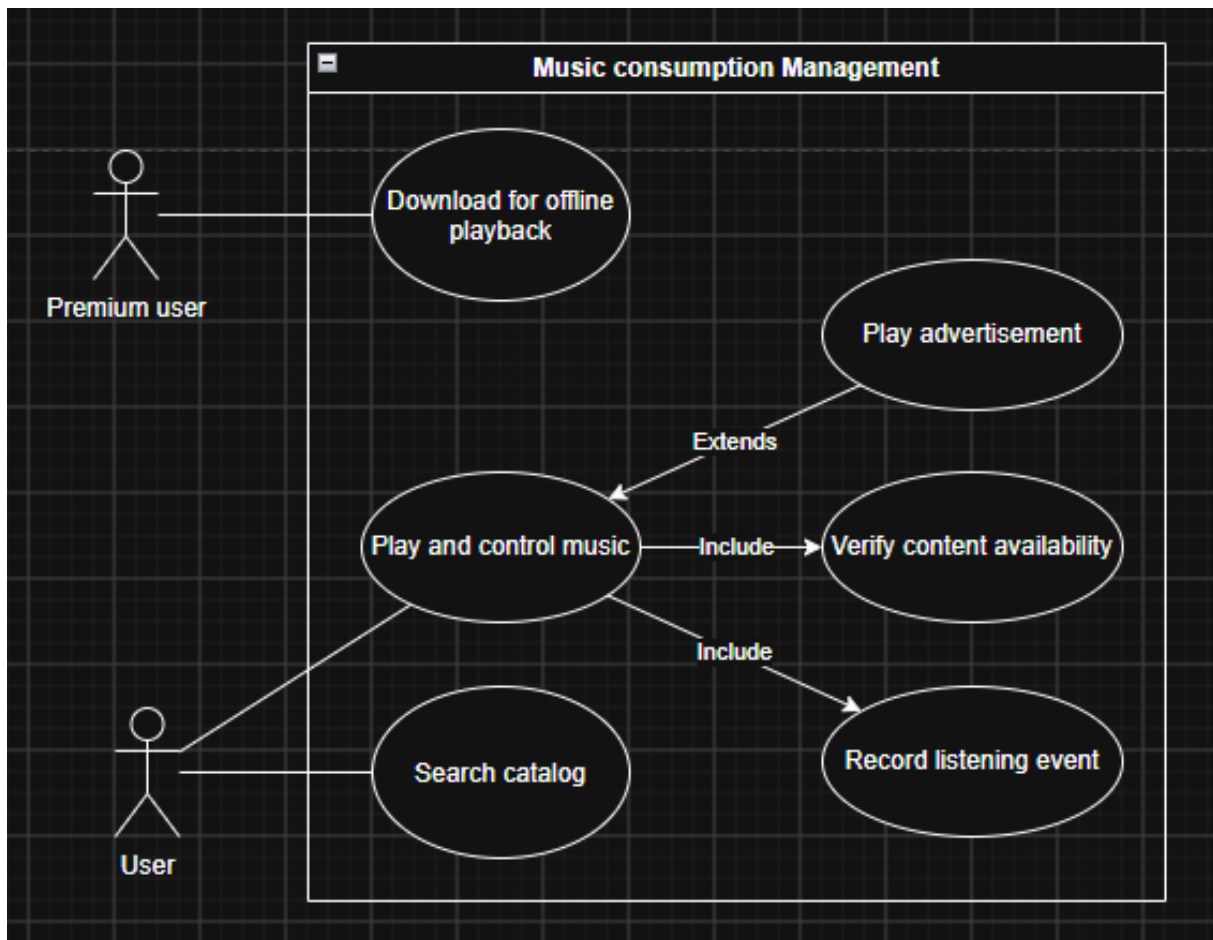


Figure 2: Use case diagram: Music Consumption Management

Use Case:	UC-MC-01 Download for Offline Playback
Actors:	Premium User
Preconditions:	The Premium User is authenticated, has an active Premium entitlement, has a network connection for the download operation, and the selected content is authorized for the user's market.
Main Flow:	1. The Premium User selects music to make available offline. 2. The system verifies that the account has an active Premium subscription. 3. The system verifies that the selected music is currently available to the user. 4. The system downloads the authorized content to the supported application environment. 5. The system associates the downloaded content with the user's offline authorization. 6. The content becomes available for offline playback.
Alternative Flows:	A1. If the subscription is not Premium or is no longer valid, the download is denied. A2. If the selected content is unavailable in the user's market, the download is not performed. A3. If the network connection is interrupted, the download remains incomplete and may be retried. A4. If the application does not reconnect to the service for more than 30 days, the offline authorization expires. The downloaded content becomes unavailable for offline playback until the user reconnects and the system successfully revalidates the Premium subscription and content authorization.
Postconditions:	Authorized music is stored in the supported playback environment and remains available for offline use only while the Premium entitlement and the most recent offline authorization remain valid.
Description:	Allows Premium Users to make authorized music available for playback without a continuous network connection. Offline authorization must be revalidated by connecting to the service at least once every 30 days; this revalidation is an offline-access rule and is not treated as the subscription billing-renewal cycle.

Table 5: Use case specification UC-MC-01: Download for Offline Playback

Use Case:	UC-MC-02 Play and Control Music
Actors:	User
Preconditions:	The user has an active application session and has selected a recording that exists in the catalog.
Main Flow:	1. The user requests playback of a music item. 2. The system executes UC-MC-04 Verify Content Availability. 3. If playback is authorized, the system initializes delivery of the selected audio. 4. The user may pause, resume, move to another track, return to a previous track, manage the playback queue, use shuffle, or use supported autoplay functionality. 5. The system keeps the current playback state synchronized with the user's session. 6. Relevant playback interactions execute UC-MC-05 Record Listening Event.
Alternative Flows:	A1. If the selected content is not available to the user, playback is denied. A2. If streaming is interrupted because of network conditions, the system attempts to recover or resume playback when possible. A3. For a Free User, UC-MC-03 Play Advertisement may extend the playback flow according to the applicable advertising policy. A4. If a requested playback control is unavailable under the current account rules, the system applies the functionality permitted by the user's plan.
Postconditions:	The playback state reflects the user's last confirmed interaction and relevant listening activity has been recorded for downstream processing.
Description:	Provides the core music-streaming functionality and allows users to control an active playback session.

Table 6: Use case specification UC-MC-02: Play and Control Music

Use Case:	UC-MC-03 Play Advertisement
Actors:	Free User
Preconditions:	A Free User has an active music-playback session and the current advertising policy determines that an advertisement may be inserted.
Main Flow:	1. During music playback, the system determines that an advertising opportunity exists. 2. The system selects or receives an advertisement that is eligible for the current session. 3. Music playback is temporarily interrupted or transitioned according to the advertising policy. 4. The advertisement is delivered to the Free User. 5. When advertisement playback finishes, the music session continues according to its previous state.
Alternative Flows:	A1. If no eligible advertisement is available, music playback continues according to the current free-tier policy. A2. If advertisement delivery fails, the system handles the failure without invalidating unrelated account or catalog functionality. A3. If the user is a Premium User, this extension is not executed.
Postconditions:	The advertising opportunity has been processed and the associated music playback session can continue.
Description:	Extends music playback for users of the free tier by inserting advertisements according to the platform's advertising rules.

Table 7: Use case specification UC-MC-03: Play Advertisement

Use Case:	UC-MC-04 Verify Content Availability
Actors:	User, indirectly through UC-MC-02 Play and Control Music
Preconditions:	A user has requested access to catalog content and the platform has availability information associated with the selected recording.
Main Flow:	1. The system identifies the requested recording. 2. The system identifies the user's applicable market and account conditions. 3. The system evaluates the rights, territories, availability periods, and relevant account restrictions associated with the recording. 4. The system determines whether the content can currently be played by the user. 5. The availability decision is returned to the playback process.
Alternative Flows:	A1. If the recording is not licensed for the user's market, access is denied. A2. If the authorized availability period has not started or has expired, access is denied. A3. If required rights information cannot be safely validated, the system does not authorize playback until a valid decision can be obtained.
Postconditions:	A valid availability decision exists for the current playback request.
Description:	Determines whether a requested recording can be accessed by the user according to rights, territory, temporal availability, and applicable account restrictions.

Table 8: Use case specification UC-MC-04: Verify Content Availability

Use Case:	UC-MC-05 Record Listening Event
Actors:	User, indirectly through UC-MC-02 Play and Control Music
Preconditions:	A playback interaction has occurred and the system has sufficient context to associate the interaction with the active user session and recording.
Main Flow:	1. The system detects a relevant playback interaction. 2. The system associates the event with the corresponding user and recording. 3. The system records relevant information such as timestamp, listening duration, market, and playback context. 4. The system validates the event structure and required information. 5. When a track has been played for at least 30 seconds, the system marks the interaction as a validated track play for the modeled listening statistics. 6. Validated playback information is stored for listening history, personalization, annual-summary functionality, and general product analytics.
Alternative Flows:	A1. If the event is incomplete or malformed, it is not treated as a validated listening event. A2. If duplicate event information is detected, the system prevents unnecessary duplicate processing. A3. If the event-processing path is temporarily unavailable, the event may remain pending until it can be processed according to the system's reliability mechanisms. A4. If playback ends before 30 seconds, the interaction may remain as raw telemetry but is not counted as a validated track play.
Postconditions:	The playback interaction has been stored with its validation state and is available to the supported downstream product features. No royalty obligation is calculated from the event within this project.
Description:	Records and validates listening interactions required to reconstruct user activity and support personalized experiences. The project keeps playback telemetry but does not implement royalty calculation or settlement.

Table 9: Use case specification UC-MC-05: Record Listening Event

Use Case:	UC-MC-06 Search Catalog
Actors:	User
Preconditions:	The user has access to the application and the music catalog is available for search operations.
Main Flow:	1. The user opens the search functionality. 2. The user enters a textual search query. 3. The system searches supported catalog metadata. 4. The system identifies matching tracks, artists, albums, and playlists. 5. Applicable visibility and availability restrictions are applied. 6. The system displays the resulting catalog matches to the user.
Alternative Flows:	A1. If no relevant catalog entries match the query, the system informs the user that no results were found. A2. Content that cannot be shown or accessed under applicable rules is excluded from the results. A3. If search functionality is temporarily degraded, the system reports that the operation cannot currently be completed.
Postconditions:	The user receives the available search results corresponding to the submitted query.
Description:	Allows users to discover content by searching the music catalog using supported metadata.

Table 10: Use case specification UC-MC-06: Search Catalog

2.3.3 Playlist Management

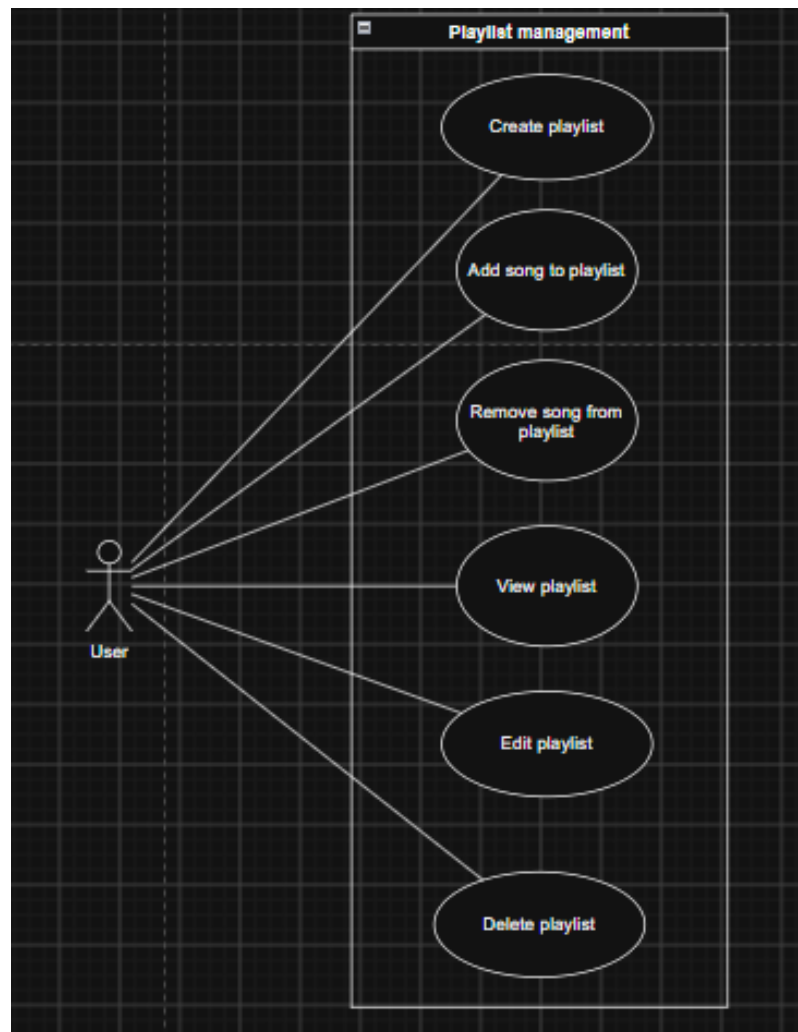


Figure 3: Use case diagram: Playlist Management

Use Case:	UC-PL-01 Create Playlist
Actors:	User
Preconditions:	The user is authenticated and has access to playlist-management functionality.
Main Flow:	1. The user selects the option to create a playlist. 2. The system creates a new playlist associated with the user's account. 3. The user provides the initial playlist information supported by the platform. 4. The system validates and stores the playlist. 5. The newly created playlist becomes available to the owner.
Alternative Flows:	A1. If required playlist information is invalid, the system requests correction before completing the creation. A2. If the operation cannot be persisted, the system reports the failure and does not present the playlist as successfully created.
Postconditions:	A new playlist associated with the user exists when the operation succeeds.
Description:	Allows users to create playlists that can later contain selected music and be managed from their account.

Table 11: Use case specification UC-PL-01: Create Playlist

Use Case:	UC-PL-02 Add Song to Playlist
Actors:	User
Preconditions:	The user is authenticated, the selected song exists in the catalog, and the user has permission to modify the target playlist.
Main Flow:	1. The user selects a song. 2. The user selects the option to add the song to a playlist. 3. The system displays the playlists that the user is permitted to modify. 4. The user selects the destination playlist. 5. The system adds the song to the playlist. 6. The updated playlist is stored and displayed.
Alternative Flows:	A1. If the user does not have permission to modify the playlist, the operation is rejected. A2. If the target playlist no longer exists, the song is not added. A3. If the modification cannot be persisted, the previous playlist content remains valid.
Postconditions:	The selected song appears in the target playlist when the operation succeeds.
Description:	Allows an authorized user to add catalog music to an existing playlist.

Table 12: Use case specification UC-PL-02: Add Song to Playlist

Use Case:	UC-PL-03 Remove Song from Playlist
Actors:	User
Preconditions:	The user is authenticated, the playlist exists, the selected song is present in the playlist, and the user is authorized to modify the playlist.
Main Flow:	1. The user opens a playlist. 2. The user selects a song contained in the playlist. 3. The user requests removal of the selected song. 4. The system verifies that the user may modify the playlist. 5. The system removes the selected song. 6. The updated playlist is stored.
Alternative Flows:	A1. If the user is not authorized to modify the playlist, the removal is denied. A2. If the selected song is no longer present in the playlist, no additional modification is performed. A3. If persistence fails, the previous confirmed playlist content is preserved.
Postconditions:	The selected song is no longer part of the playlist when the removal succeeds.
Description:	Allows an authorized user to remove music from a playlist.

Table 13: Use case specification UC-PL-03: Remove Song from Playlist

Use Case:	UC-PL-04 View Playlist
Actors:	User
Preconditions:	The requested playlist exists and the user is allowed to view it according to its current visibility and access configuration.
Main Flow:	1. The user selects a playlist. 2. The system retrieves the playlist information. 3. The system retrieves the songs currently associated with the playlist. 4. The system applies the relevant visibility and availability rules. 5. The playlist and its accessible contents are displayed to the user.
Alternative Flows:	A1. If the playlist does not exist, the system informs the user. A2. If the playlist is not accessible to the requesting user, the system denies access. A3. Music that is temporarily unavailable may remain represented according to the platform's supported unavailable-content behavior but cannot be played.
Postconditions:	The user has access to the current view of the requested playlist when access is authorized.
Description:	Allows users to inspect a playlist and the music currently associated with it.

Table 14: Use case specification UC-PL-04: View Playlist

Use Case:	UC-PL-05 Edit Playlist
Actors:	User
Preconditions:	The user is authenticated, the playlist exists, and the user has permission to edit the playlist.
Main Flow:	1. The user opens a playlist for editing. 2. The system retrieves the current playlist configuration. 3. The user modifies supported playlist information, such as its name, description, visibility, image, track ordering, or collaboration configuration. 4. The system validates the requested changes. 5. The system stores the confirmed modifications. 6. The updated playlist is displayed.
Alternative Flows:	A1. If the user does not have editing permission, the operation is denied. A2. If one or more values are invalid, the system requests correction. A3. If concurrent modifications occur, the system preserves confirmed changes according to the playlist-consistency rules and returns the resulting playlist state.
Postconditions:	The playlist contains the accepted modifications and remains associated with its authorized users.
Description:	Allows an authorized user to modify playlist information and supported playlist configuration.

Table 15: Use case specification UC-PL-05: Edit Playlist

Use Case:	UC-PL-06 Delete Playlist
Actors:	User
Preconditions:	The user is authenticated, the playlist exists, and the user has permission to delete it.
Main Flow:	1. The user selects a playlist that they are authorized to delete. 2. The user requests deletion. 3. The system verifies the user's permission. 4. The system removes the playlist from the owner's active playlist collection. 5. The system confirms successful deletion.
Alternative Flows:	A1. If the user is not authorized to delete the playlist, the request is rejected. A2. If the playlist has already been deleted, no additional deletion operation is required. A3. If deletion cannot be completed, the previously confirmed playlist remains available.
Postconditions:	The deleted playlist is no longer available as an active playlist of the owner.
Description:	Allows an authorized user to remove a playlist that they control.

Table 16: Use case specification UC-PL-06: Delete Playlist

2.3.4 Library Management

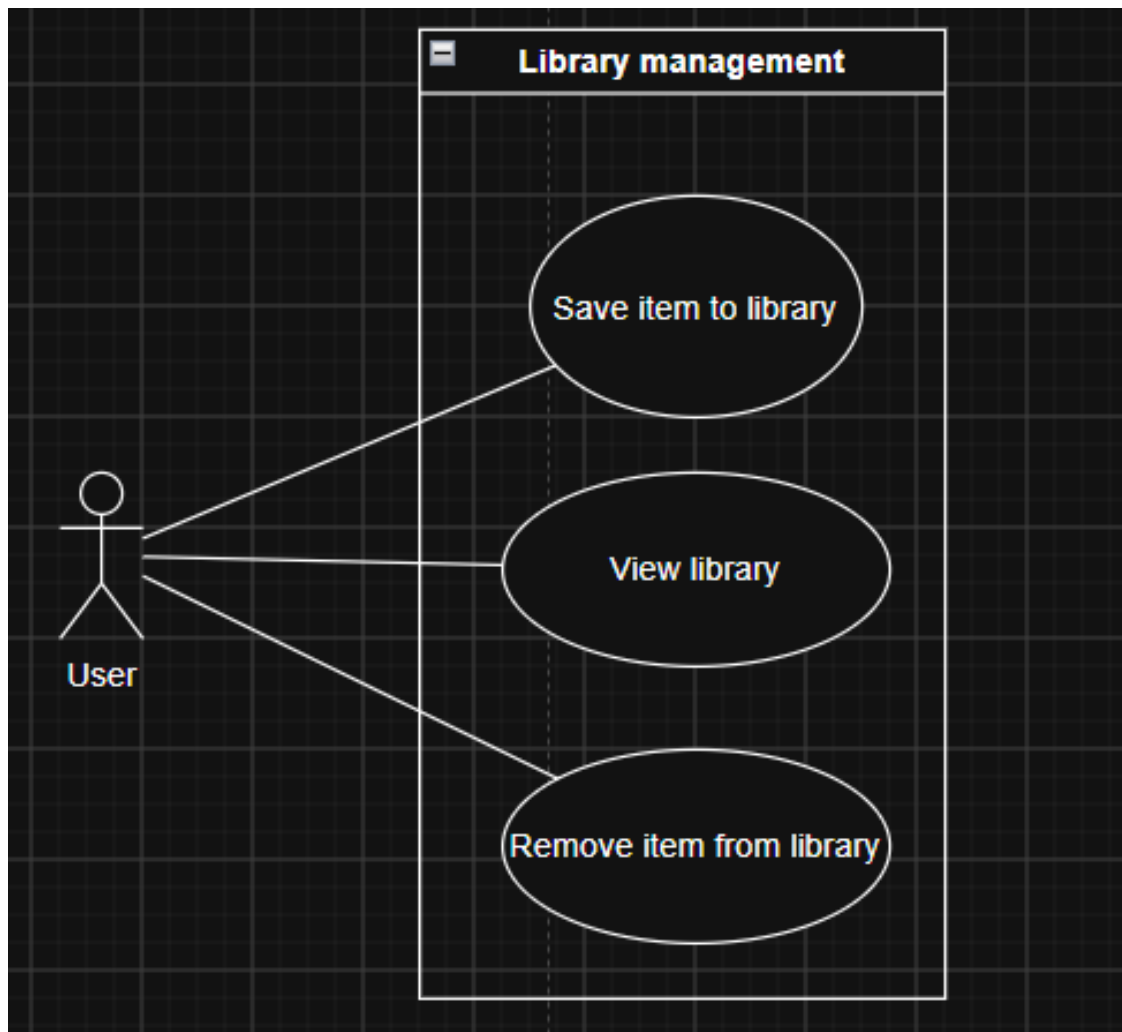


Figure 4: Use case diagram: Library Management

Use Case:	UC-LI-01 Save Item to Library
Actors:	User
Preconditions:	The user is authenticated and the selected track, album, or playlist exists and can be accessed by the user.
Main Flow:	1. The user selects a track, album, or playlist. 2. The user requests that the item be saved to the personal library. 3. The system verifies the selected item. 4. The system associates the item with the user's library. 5. The updated library state is stored. 6. The system confirms that the item is saved.
Alternative Flows:	A1. If the item is already saved, the system does not create an unnecessary duplicate association. A2. If the item no longer exists or cannot be accessed, the operation is not completed. A3. If persistence fails, the previous library state is preserved.
Postconditions:	The selected item is associated with the user's personal library.
Description:	Allows users to save supported catalog items so they can be accessed later from their personal library.

Table 17: Use case specification UC-LI-01: Save Item to Library

Use Case:	UC-LI-02 View Library
Actors:	User
Preconditions:	The user is authenticated and has access to their personal library.
Main Flow:	1. The user opens the personal library. 2. The system retrieves the items currently associated with the user's library. 3. The system organizes the supported saved content categories. 4. Current availability information is applied where necessary. 5. The system displays the user's saved tracks, albums, and playlists.
Alternative Flows:	A1. If the library contains no saved items, the system displays an empty-library state. A2. If an item saved previously is no longer playable, the platform applies the corresponding unavailable-content behavior. A3. If part of the library cannot be retrieved, the system may display the information that remains available.
Postconditions:	The current library information has been presented to the user without changing the saved content.
Description:	Allows a user to consult the tracks, albums, and playlists stored in their personal library.

Table 18: Use case specification UC-LI-02: View Library

Use Case:	UC-LI-03 Remove Item from Library
Actors:	User
Preconditions:	The user is authenticated and the selected track, album, or playlist is currently saved in the user's library.
Main Flow:	1. The user selects an item in the personal library. 2. The user requests that the item be removed. 3. The system verifies that the association exists. 4. The system removes the association between the item and the user's library. 5. The updated library state is stored. 6. The system confirms the removal.
Alternative Flows:	A1. If the item has already been removed, no additional modification is required. A2. If the modification cannot be persisted, the previous confirmed library state remains valid.
Postconditions:	The selected item is no longer saved in the user's personal library.
Description:	Allows users to remove previously saved tracks, albums, or playlists from their personal library.

Table 19: Use case specification UC-LI-03: Remove Item from Library

2.3.5 Personalization and Listening Insights Management

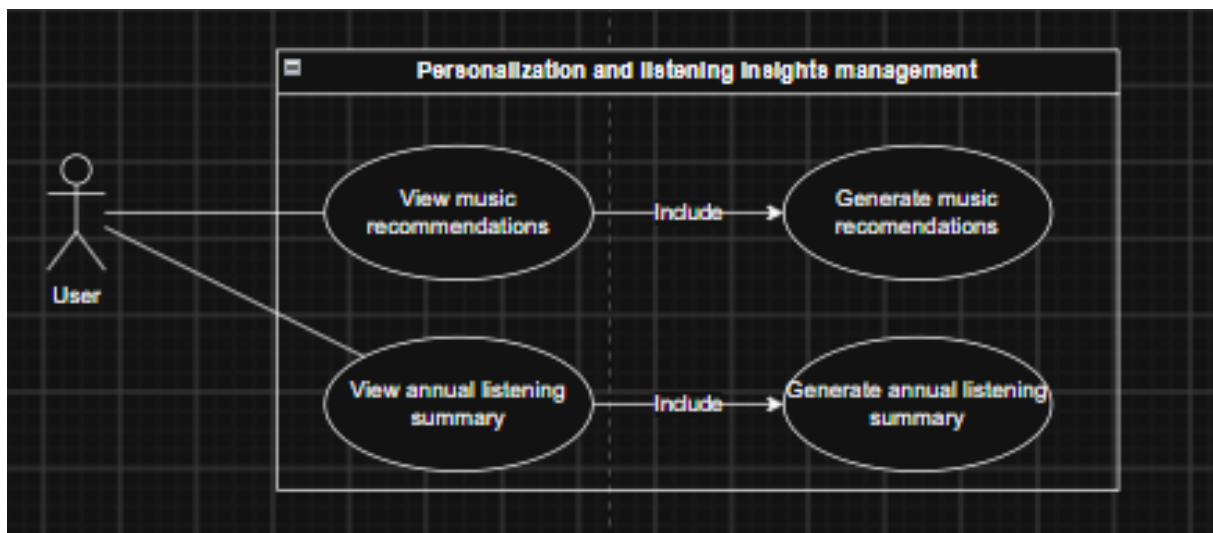


Figure 5: Use case diagram: Personalization and Listening Insights Management

Use Case:	UC-PI-01 View Music Recommendations
Actors:	User
Preconditions:	The user has access to the platform and the recommendation functionality is available.
Main Flow:	1. The user opens a surface that presents music recommendations. 2. The system executes UC-PI-02 Generate Music Recommendations. 3. The generated recommendation set is retrieved. 4. The system applies current content-availability restrictions. 5. The recommendations are displayed to the user. 6. The user may select recommended content for subsequent playback.
Alternative Flows:	A1. If personalization data is insufficient, the system displays the fallback recommendations generated by UC-PI-02. A2. If some recommended content becomes unavailable, that content is removed or replaced before presentation. A3. If recommendation functionality is temporarily unavailable, basic catalog and playback functionality remain accessible.
Postconditions:	A recommendation set has been presented to the user without modifying the underlying catalog.
Description:	Allows users to discover music selected by the platform according to available personalization information.

Table 20: Use case specification UC-PI-01: View Music Recommendations

Use Case:	UC-PI-02 Generate Music Recommendations
Actors:	User, indirectly through UC-PI-01 View Music Recommendations
Preconditions:	A recommendation request exists and the system can access the available user profile, listening, library, or catalog information required by the recommendation process.
Main Flow:	1. The system receives a request to generate recommendations for the user. 2. The system retrieves available preference signals, including listening behavior, liked or saved content, genres, artists, and recent activity. 3. The system processes the available information using the supported recommendation mechanisms. 4. Candidate music is ranked according to the calculated relevance. 5. Unavailable or ineligible content is excluded. 6. The resulting recommendation set is returned to the requesting experience.
Alternative Flows:	A1. If insufficient user information exists, the system generates fallback recommendations such as popular or regionally relevant music. A2. If some personalization signals are temporarily unavailable, the system uses the information that remains valid. A3. If recommendation generation fails, no recommendation result is returned, but unrelated playback functionality remains operational.
Postconditions:	A recommendation set appropriate to the information available for the user has been produced.
Description:	Generates personalized or fallback music recommendations that can later be presented to the user.

Table 21: Use case specification UC-PI-02: Generate Music Recommendations

Use Case:	UC-PI-03 View Annual Listening Summary
Actors:	User
Preconditions:	The annual listening-summary experience is available for the corresponding period and the user has an account eligible to access the experience.
Main Flow:	1. The user opens the annual listening-summary experience. 2. The system executes or retrieves the result of UC-PI-04 Generate Annual Listening Summary. 3. The system loads the annual statistics available for the user. 4. The information is organized into the supported annual-summary presentation. 5. The personalized summary is displayed to the user.
Alternative Flows:	A1. If only part of the supported summary is available, the system presents the subset that can be generated from eligible data. A2. If the user does not satisfy the minimum data conditions required for the experience, the system displays the supported fallback behavior. A3. If annual-summary functionality is unavailable, normal music streaming remains unaffected.
Postconditions:	The user has viewed the annual listening information available for their account.
Description:	Allows users to access a personalized yearly overview derived from their eligible music-listening activity.

Table 22: Use case specification UC-PI-03: View Annual Listening Summary

Use Case:	UC-PI-04 Generate Annual Listening Summary
Actors:	User, indirectly through UC-PI-03 View Annual Listening Summary
Preconditions:	Eligible listening events exist for the user for the annual analysis period and the annual-summary generation process is available.
Main Flow:	1. The system identifies the user and the applicable annual analysis period. 2. The system retrieves the eligible listening history associated with the user. 3. The system aggregates supported statistics such as listening activity, tracks, artists, or other music-consumption indicators. 4. The system applies the rules required by the annual-summary experience. 5. The personalized annual-summary information is generated. 6. The generated information becomes available to the user-facing annual-summary experience.
Alternative Flows:	A1. If insufficient eligible listening information exists, the system generates only the supported subset or fallback experience. A2. Invalid listening events are excluded from the annual aggregation. A3. If generation cannot be completed, the summary remains unavailable until a valid result can be produced.
Postconditions:	A personalized annual listening summary or supported reduced version is available for presentation to the user.
Description:	Processes eligible listening history to create the personalized annual listening-summary experience.

Table 23: Use case specification UC-PI-04: Generate Annual Listening Summary

2.3.6 Release Management

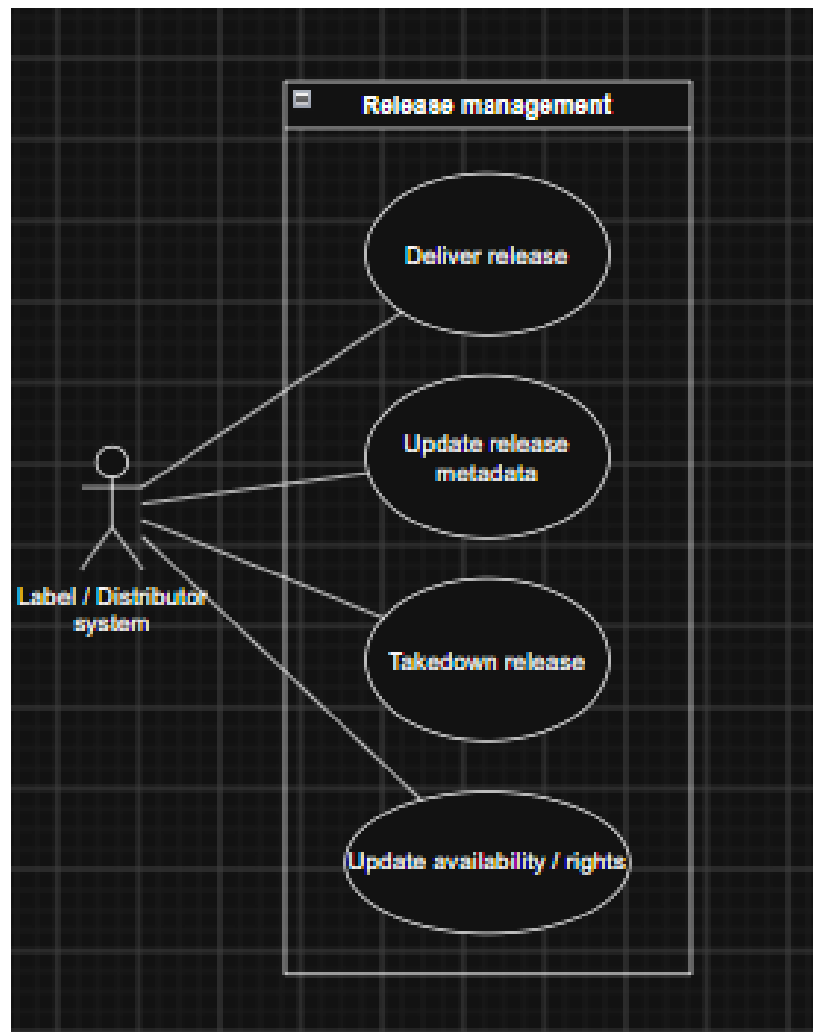


Figure 6: Use case diagram: Release Management

Use Case:	UC-RM-01 Deliver Release
Actors:	Label / Distributor System
Preconditions:	The external Label / Distributor System is authorized to deliver content and has prepared a release containing the required music assets and metadata.
Main Flow:	1. The Label / Distributor System sends a release to the platform. 2. The platform receives the release information, recordings, and associated metadata. 3. The platform validates the structure and required information. 4. The platform associates the recordings with the corresponding release and participants. 5. The platform stores the valid release information. 6. The release becomes eligible for catalog publication according to its availability information.
Alternative Flows:	A1. If mandatory data is missing or invalid, the delivery is rejected or held until corrected. A2. If identifiers conflict with existing catalog information, the system prevents creation of an inconsistent duplicate. A3. If ingestion fails temporarily, the release is not considered successfully incorporated until processing completes.
Postconditions:	A valid release and its associated information exist in the platform catalog or the invalid delivery has been rejected.
Description:	Allows an authorized label or distributor system to deliver new music releases and the information required to incorporate them into the Spotify catalog.

Table 24: Use case specification UC-RM-01: Deliver Release

Use Case:	UC-RM-02 Update Release Metadata
Actors:	Label / Distributor System
Preconditions:	The Label / Distributor System is authorized to modify the release and the corresponding release already exists in the platform.
Main Flow:	1. The external system identifies the release to be updated. 2. The external system sends the updated metadata. 3. The platform identifies the existing release. 4. The platform validates the new information. 5. The accepted metadata replaces or supplements the corresponding previous values. 6. The catalog reflects the confirmed metadata update.
Alternative Flows:	A1. If the release cannot be found, the update is rejected. A2. If the external system is not authorized to modify the release, the operation is rejected. A3. If the supplied metadata is invalid, the previous valid metadata is preserved.
Postconditions:	The release contains the latest validated metadata supplied by the authorized content provider.
Description:	Allows an authorized label or distributor system to correct or update metadata associated with music already delivered to the platform.

Table 25: Use case specification UC-RM-02: Update Release Metadata

Use Case:	UC-RM-03 Takedown Release
Actors:	Label / Distributor System
Preconditions:	The release exists in the catalog and the requesting Label / Distributor System is authorized to request its takedown.
Main Flow:	1. The external system identifies the release that must be taken down. 2. The external system submits the takedown request. 3. The platform validates the release identifier and the provider's authority. 4. The platform marks the release as inactive or unavailable for the affected catalog scope instead of physically deleting its catalog record. 5. Playback and discovery behavior are updated according to the takedown. 6. The platform preserves the release identifiers, metadata, provider associations, and historical state required for traceability and future reconciliation or reactivation.
Alternative Flows:	A1. If the release does not exist, no takedown modification is performed. A2. If the requesting system is not authorized, the request is rejected. A3. If the takedown applies only to certain territories or periods, the platform updates only the affected availability scope instead of applying a global removal.
Postconditions:	The release remains stored as a historical catalog record but is unavailable for playback and discovery according to the confirmed takedown scope.
Description:	Allows an authorized content provider to make a previously delivered release unavailable without hard-deleting the underlying catalog record. This preserves metadata and identifiers required for traceability, reconciliation, and possible future reactivation.

Table 26: Use case specification UC-RM-03: Takedown Release

Use Case:	UC-RM-04 Update Availability / Rights
Actors:	Label / Distributor System
Preconditions:	The relevant release or recording exists in the catalog and the external system is authorized to supply rights and availability information for it.
Main Flow:	1. The external system identifies the affected release or recording. 2. The external system submits updated rights, territory, or availability-period information. 3. The platform validates the supplied information. 4. The platform determines the catalog scope affected by the update. 5. The platform modifies the availability rules for the corresponding content. 6. Future playback requests use the updated rights information.
Alternative Flows:	A1. If the update is invalid, the previous confirmed rights information remains active. A2. If authorization for a territory expires, the content becomes unavailable in that territory. A3. If a future availability date is supplied, the release remains unavailable until the authorized period begins.
Postconditions:	The catalog availability rules reflect the latest validated rights and territorial information.
Description:	Allows authorized content providers to modify when, where, and under which rights conditions music can be made available to users.

Table 27: Use case specification UC-RM-04: Update Availability / Rights

2.3.7 Subscription Management

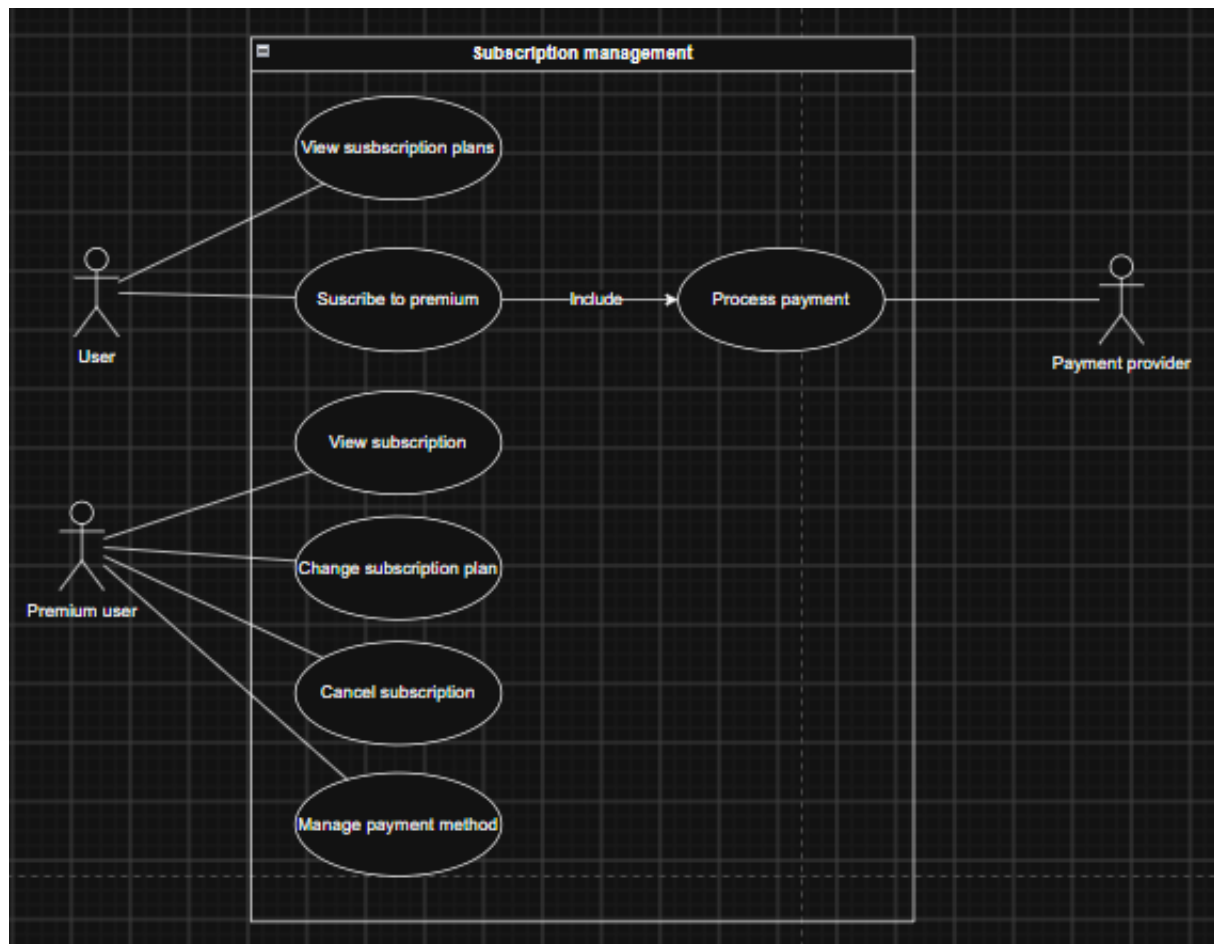


Figure 7: Use case diagram: Subscription Management

Use Case:	UC-SU-01 View Subscription Plans
Actors:	User
Preconditions:	The user has access to the subscription-management functionality and subscription-plan information is available.
Main Flow:	1. The user opens the subscription section. 2. The system retrieves the subscription plans available to the user. 3. The system displays the applicable plan information and supported features. 4. The user may inspect the available plans and select one for a subsequent subscription operation.
Alternative Flows:	A1. Plans that are not available for the user's market or eligibility conditions are not offered as selectable plans. A2. If subscription information cannot be retrieved, the system informs the user that plans cannot currently be displayed.
Postconditions:	The user has received the currently applicable subscription-plan information; no subscription state is changed.
Description:	Allows users to inspect the Premium subscription alternatives available to their account.

Table 28: Use case specification UC-SU-01: View Subscription Plans

Use Case:	UC-SU-02 Subscribe to Premium
Actors:	User; Payment Provider as secondary actor through UC-SU-03 Process Payment
Preconditions:	The user has a registered account, has selected an available Premium plan, and the required payment operation can be initiated.
Main Flow:	1. The user selects an available Premium subscription plan. 2. The system presents the subscription conditions associated with the selected plan. 3. The user confirms the subscription request. 4. The system executes UC-SU-03 Process Payment. 5. The payment operation is successfully confirmed. 6. The system updates the account to the corresponding Premium subscription state. 7. Premium functionality becomes available according to the plan.
Alternative Flows:	A1. If payment is rejected, the subscription is not activated. A2. If the selected plan is no longer available, the user must choose another applicable plan. A3. If payment confirmation is temporarily pending, Premium access is not granted until the required confirmation is obtained. A4. If the user cancels the operation before confirmation, the existing subscription state remains unchanged.
Postconditions:	The user has an active Premium subscription when payment and subscription activation succeed. Otherwise, the previous subscription state is preserved.
Description:	Allows a user to obtain Premium access by selecting a subscription plan and successfully completing the required payment process.

Table 29: Use case specification UC-SU-02: Subscribe to Premium

Use Case:	UC-SU-03 Process Payment
Actors:	Payment Provider
Preconditions:	A subscription-related payment operation has been initiated and the platform has the information required to request payment processing from the external Payment Provider.
Main Flow:	1. The platform creates the payment request associated with the subscription operation. 2. The platform sends the required transaction information to the Payment Provider. 3. The Payment Provider processes the operation. 4. The Payment Provider returns the payment result. 5. The platform validates the returned result. 6. The confirmed payment result is made available to the corresponding subscription process.
Alternative Flows:	A1. If the Payment Provider rejects the transaction, the platform records the failed result and does not activate the requested paid entitlement. A2. If the provider does not return a definitive result, the operation remains pending until its state can be reconciled. A3. If communication with the Payment Provider fails, the platform must not assume that payment succeeded.
Postconditions:	A confirmed, rejected, or pending payment result exists and can be used to determine the corresponding subscription state.
Description:	Represents the integration between the platform and the external payment system required to process subscription-related payments.

Table 30: Use case specification UC-SU-03: Process Payment

Use Case:	UC-SU-04 View Subscription
Actors:	Premium User
Preconditions:	The Premium User is authenticated and has subscription information associated with the account.
Main Flow:	1. The Premium User opens subscription management. 2. The system identifies the subscription associated with the account. 3. The system retrieves the current plan and its relevant state. 4. The system displays the subscription information to the user.
Alternative Flows:	A1. If subscription information is temporarily unavailable, the system informs the user that the current subscription state cannot be displayed. A2. If the subscription is scheduled for cancellation or another pending change, the system displays the corresponding confirmed state.
Postconditions:	The subscription information has been presented without modifying the current subscription.
Description:	Allows a Premium User to consult the current state and plan associated with their subscription.

Table 31: Use case specification UC-SU-04: View Subscription

Use Case:	UC-SU-05 Change Subscription Plan
Actors:	Premium User
Preconditions:	The Premium User is authenticated, has an active Premium subscription, and at least one alternative plan is available to the account.
Main Flow:	1. The Premium User opens subscription management. 2. The system displays the plans available for migration. 3. The user selects a new subscription plan. 4. The system determines the effects and effective conditions of the plan change. 5. The user confirms the modification. 6. When required, the corresponding payment operation is processed. 7. The system updates the subscription according to the confirmed plan change.
Alternative Flows:	A1. If the selected plan is not available to the user, the change is rejected. A2. If an additional required payment operation fails, the previous plan remains active according to its current validity. A3. If the change is scheduled for a future billing period, the system preserves the current plan until the corresponding effective date.
Postconditions:	The subscription reflects the confirmed new plan immediately or records the future plan transition according to the applicable subscription rules.
Description:	Allows a Premium User to migrate from the current subscription plan to another plan available to the account.

Table 32: Use case specification UC-SU-05: Change Subscription Plan

Use Case:	UC-SU-06 Cancel Subscription
Actors:	Premium User
Preconditions:	The Premium User is authenticated and has an active or currently valid Premium subscription that can be cancelled.
Main Flow:	1. The Premium User opens subscription management. 2. The user selects the option to cancel the Premium subscription. 3. The system presents the effect of the cancellation. 4. The user confirms the cancellation. 5. The system records the cancellation. 6. The platform determines the remaining period for which Premium entitlements continue to be valid. 7. When the paid period ends, the account transitions to the corresponding Free state.
Alternative Flows:	A1. If the user abandons the cancellation before confirmation, the subscription remains unchanged. A2. If the subscription is already scheduled for cancellation, the system displays the existing cancellation state. A3. If the cancellation cannot be processed, the existing subscription state remains unchanged until a valid cancellation is confirmed.
Postconditions:	The Premium subscription is scheduled to terminate according to its remaining validity period, after which the account loses the corresponding Premium entitlements.
Description:	Allows a Premium User to cancel future continuation of the paid subscription without necessarily removing already-paid access immediately.

Table 33: Use case specification UC-SU-06: Cancel Subscription

Use Case:	UC-SU-07 Manage Payment Method
Actors:	Premium User; Payment Provider when external validation is required
Preconditions:	The Premium User is authenticated and has access to subscription and payment management.
Main Flow:	1. The Premium User opens payment-method management. 2. The system displays the supported payment information associated with the subscription. 3. The user requests to add, replace, or update the payment method. 4. The system collects the required payment information through the supported secure payment flow. 5. When required, the information is validated through the external Payment Provider. 6. The system associates the confirmed payment method with future subscription operations. 7. The system confirms the update to the user.
Alternative Flows:	A1. If the Payment Provider rejects the supplied payment method, the new method is not activated. A2. If required payment information is incomplete, the system requests correction. A3. If external validation is temporarily unavailable, the system does not mark the new payment method as confirmed. A4. If the update fails, the previously confirmed payment configuration remains unchanged.
Postconditions:	The subscription uses the newly confirmed payment method for applicable future operations, or the previous confirmed payment configuration remains active.
Description:	Allows a Premium User to manage the payment method associated with the subscription while delegating sensitive payment processing to the external Payment Provider.

Table 34: Use case specification UC-SU-07: Manage Payment Method

2.4 System Boundaries

Inside the System Scope	Outside the System Scope
Music releases, recordings, metadata, catalog availability, and soft takedown state management.	Artistic production and recording of musical works, and legal negotiation of licensing contracts between Spotify and rights holders.
User registration, authentication, account management, and subscription-state management for multiple Premium plan types.	Internal banking or card-network processing, payment authorization/capture, refund processing, fraud/risk decisions, and financial settlement performed by the external Payment Provider or financial institutions.

Inside the System Scope	Outside the System Scope
Payment orchestration required to request a subscription-related transaction, receive its result, and update the corresponding subscription entitlement.	Storage or direct processing of raw card or bank-account credentials by the modeled platform.
Music playback, playback-session control, search, library, playlists, recommendations, and Premium offline playback with periodic authorization revalidation.	Physical infrastructure of user devices and telecommunications networks.
Playback telemetry, validated track-play classification, listening history, and annual listening summaries.	Royalty calculation, royalty settlement, royalty reporting, and distribution of money to labels, distributors, publishers, collecting societies, artists, or composers.
Free-tier advertising opportunities and advertisement insertion into music playback through the supported advertising integration.	Advertiser campaign creation, targeting-engine internals, ad auctions or bidding, advertiser billing, and advertising financial settlement.
Rights, territory, and availability-period updates supplied by authorized content providers.	Podcast and audiobook catalog management, and real-time Spotify Jam / group listening sessions.

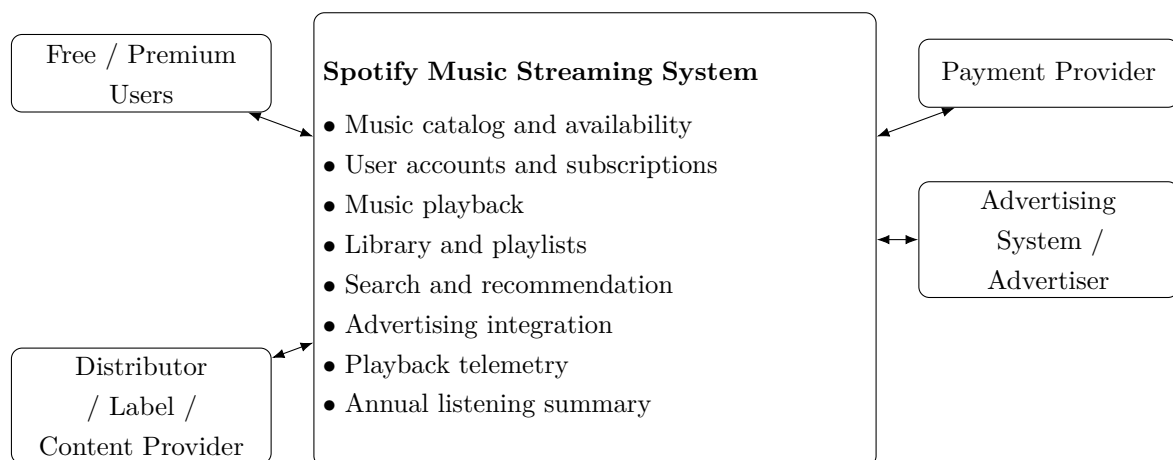


Figure 8: System boundary for the modeled Spotify music-streaming platform

3 Functional Requirements

The following twelve functional requirements define the principal behaviors included in the scope of the modeled music-streaming system. Closely related operations have been consolidated to keep the specification within the desired level of abstraction while preserving traceability to the use cases.

3.1 Music Catalog Management

Requirement ID:	FR-MM-01
Requirement Name:	Manage Music Catalog and Availability
Requirement Description:	The system shall receive music releases, recordings, and metadata from authorized distributors, labels, or content providers; validate and associate the received information; update release metadata; and apply rights, territory, and availability-period information to determine where and when content can be played. When an authorized takedown is received, the system shall mark the affected release or recording as inactive or unavailable for the corresponding scope instead of hard-deleting the catalog record, preserving its identifiers, metadata, provider associations, and historical state for traceability, reconciliation, and possible future reactivation.
Bibliographic Reference:	Spotify for Artists [1]; DDEX Release Delivery / ERN [2].

Table 36: Functional requirement FR-MM-01

3.2 Users and Accounts

Requirement ID:	FR-US-01
Requirement Name:	Manage User Account
Requirement Description:	The system shall allow a guest to register an account and shall allow a registered user to authenticate, access the account, and modify the supported account information and settings. Invalid registration or authentication data shall not create or authenticate an account.
Bibliographic Reference:	System-defined requirement.

Table 37: Functional requirement FR-US-01

3.3 Subscriptions and Payments

Requirement ID:	FR-SU-01
Requirement Name:	Manage Premium Subscription and Payment Integration
Requirement Description:	The system shall support multiple Premium plan types and shall allow users to view available plans, subscribe to an eligible plan, view the current subscription, change plan, cancel future renewal, and manage the payment method reference associated with the subscription. For payment-dependent operations, the platform shall send the required transaction request to an external Payment Provider, receive the confirmed, rejected, or pending result, and update the subscription state and entitlements accordingly. Sensitive card or bank-account processing, payment authorization/capture, refunds, fraud/risk decisions, and financial settlement are outside the modeled system and remain the responsibility of the external payment ecosystem.
Bibliographic Reference:	Spotify Support — Cancel Premium [10]; system-defined scope for the external payment integration.

Table 38: Functional requirement FR-SU-01

3.4 Offline Playback

Requirement ID:	FR-US-02
Requirement Name:	Download Music for Offline Playback
Requirement Description:	The system shall allow Premium users to download authorized music for offline playback inside the supported application environment. The offline authorization shall be revalidated by connecting to the service at least once every 30 days. If the user does not reconnect within that period, or if the Premium entitlement or content authorization is no longer valid, the downloaded content shall become unavailable for offline playback until authorization is successfully revalidated. The 30-day revalidation rule is independent of the subscription billing-renewal cycle.
Bibliographic Reference:	Spotify Support — Listen Offline [3].

Table 39: Functional requirement FR-US-02

3.5 Library

Requirement ID:	FR-LI-01
Requirement Name:	Manage Personal Library
Requirement Description:	The system shall allow authenticated users to save tracks, albums, and playlists to their personal library, view the current library, and remove previously saved items without creating duplicate library associations.
Bibliographic Reference:	Spotify Support — Your Library [4].

Table 40: Functional requirement FR-LI-01

3.6 Playlists

Requirement ID:	FR-PL-01
Requirement Name:	Manage Playlists
Requirement Description:	The system shall allow authenticated users to create, view, edit, and delete playlists and to add, remove, and reorder tracks when they are authorized to modify the target playlist. Supported playlist properties may include name, description, image, visibility, ordering, and collaboration configuration. Unauthorized modifications shall be rejected and confirmed concurrent changes shall not be silently lost.
Bibliographic Reference:	Spotify Support — Playlist Privacy and Access [5]; Spotify Support — Collaborative Playlists [6].

Table 41: Functional requirement FR-PL-01

3.7 Search

Requirement ID:	FR-SE-01
Requirement Name:	Search Music Catalog
Requirement Description:	The system shall allow users to perform textual queries over the music catalog and shall return matching tracks, artists, albums, and playlists using supported metadata. Search results shall respect applicable visibility and content availability restrictions.
Bibliographic Reference:	System-defined requirement based on the modeled catalog and availability rules.

Table 42: Functional requirement FR-SE-01

3.8 Playback

Requirement ID:	FR-PB-01
Requirement Name:	Play and Control Music
Requirement Description:	The system shall allow a user to request and control music playback. Before playback begins, the system shall verify the requested recording against the applicable rights, territory, availability period, account, and plan rules. For authorized playback, the user shall be able to use the supported controls, including pause, resume, next or previous track, queue management, autoplay, and shuffle. The system shall preserve the relevant playback state associated with the user's account so that a later supported session can recover it.
Bibliographic Reference:	DDEX Release Delivery / ERN [2]; system-defined playback control behavior.

Table 43: Functional requirement FR-PB-01

3.9 Music Recommendation

Requirement ID:	FR-RE-01
Requirement Name:	Recommend Music
Requirement Description:	The system shall generate and present music recommendations using available signals such as listening behavior, saved or liked content, genres, artists, and recent activity. If insufficient information exists for personalized recommendations, the system shall provide a supported fallback such as popular or regionally relevant music. Unavailable content shall not be presented as a playable recommendation.
Bibliographic Reference:	Spotify Engineering — Humans + Machines: A Look Behind Spotify's Algorithmic Playlists [7].

Table 44: Functional requirement FR-RE-01

3.10 Advertising Management

Requirement ID:	FR-AD-01
Requirement Name:	Insert Advertising for Free-Tier Playback
Requirement Description:	The system shall handle advertising opportunities during Free User music playback according to the current advertising policy. When an advertising opportunity exists, the platform shall request or receive an eligible advertisement through the supported external advertising integration, deliver it within the playback flow, and resume music playback afterward. Campaign creation, advertiser account management, targeting-engine internals, auctions or bidding, advertiser billing, and advertising financial settlement are outside the modeled system.
Bibliographic Reference:	Spotify Advertising [8, 9].

Table 45: Functional requirement FR-AD-01

3.11 Telemetry

Requirement ID:	FR-TE-01
Requirement Name:	Register and Validate Playback Events
Requirement Description:	The system shall register playback interactions including the recording, listening duration, timestamp, market, and playback context. For the modeled validated track-play counter, a track shall be counted after at least 30 seconds of playback. Shorter interactions may remain as raw telemetry but shall not be classified as validated track plays. Validated listening information may be used for listening history, personalization, annual listening summaries, and general product analytics. The project shall not calculate royalty obligations, royalty settlement amounts, or payments from these events.
Bibliographic Reference:	Spotify Support — How Your Streams Are Counted [12]; Spotify Engineering — Data Platform Explained Part II [15].

Table 46: Functional requirement FR-TE-01

3.12 Annual Listening Summary

Requirement ID:	FR-WR-01
Requirement Name:	Generate Annual Listening Summary
Requirement Description:	The system shall use the user's eligible and validated listening history to generate and present a personalized annual listening summary containing the supported statistics about the user's music consumption. If insufficient data exists, the system shall provide the supported reduced or fallback experience without affecting normal music playback.
Bibliographic Reference:	Spotify Newsroom — Wrapped Methodology [11].

Table 47: Functional requirement FR-WR-01

4 Non-Functional Requirements

The following requirements define measurable operational constraints for the modeled system. Official Spotify data is used whenever such data is publicly available. Where Spotify does not disclose an exact production target, the value is explicitly identified as a proposed system requirement or derived calculation rather than an official Spotify Service Level Objective.

4.1 Scale and Expected Users

As of Q2 2026, Spotify reported 777 million Monthly Active Users (MAUs) and 300 million Premium Subscribers [19]. Spotify's official Q2 2026 filing reported 494 million Ad-Supported MAUs [20].

Spotify Engineering subsequently reported approximately 100 million concurrent clients at any given moment [16]. Spotify does not publicly disclose its total number of registered accounts, so MAUs are used as the operational user-base benchmark.

Spotify Investor Relations reports more than 100 million music tracks and availability in 184 markets [21].

ID	Requirement / Metric	Target / Benchmark	Source / Justification
NFR-SC-01	The system shall support a user population comparable to Spotify’s current operational scale.	777 M MAUs; 300 M Premium; 494 M Ad-Supported MAUs.	Spotify Q2 2026 reporting [19, 20].
NFR-SC-02	The system shall support at least the publicly reported concurrent-client scale.	Approximately 100 M concurrent clients.	Spotify Engineering reports around 100 million concurrent clients at any given moment [16].
NFR-SC-03	The music catalog shall support at least the publicly reported Spotify music catalog scale.	More than 100 M music tracks.	Spotify Investor Relations [21].
NFR-SC-04	The system shall support the geographic scope currently reported by Spotify.	184 markets.	Spotify Investor Relations [21].

Table 48: Scale and expected-user non-functional requirements

4.2 Throughput and Request Volume

Spotify Engineering reports that the platform currently processes between 11 and 12 million backend requests per second [16].

This corresponds to:

$$11 \times 10^6 \times 3600 = 39.6 \times 10^9$$

and

$$12 \times 10^6 \times 3600 = 43.2 \times 10^9$$

backend requests per hour.

Using the midpoint of the reported range, the modeled normal hourly workload is:

$$11.5 \times 10^6 \times 3600 = 41.4 \times 10^9$$

requests per hour.

Spotify also reported that its global user base generates approximately 3.4 trillion events and taste signals every day across all product verticals and surfaces [22]. Because this measurement is not limited exclusively to music, it is used as a conservative platform-scale reference rather than as a music-only event count.

The corresponding average signal rate is:

$$\frac{3.4 \times 10^{12}}{86,400} \approx 39.35 \times 10^6$$

events and taste signals per second.

For high-demand campaigns, Spotify Engineering reports that the first three to four hours after Wrapped launch produce some of the campaign’s most demanding traffic. The campaign’s principal service was load-tested for traffic on the order of tens of thousands of requests per second across major regions [17].

Spotify additionally reports that more than 100,000 new songs are released every day in the global music ecosystem [23]. This value does not mean that all of those songs are uploaded to Spotify, but it provides a useful reference for continuous catalog-ingestion pressure.

ID	Requirement	Target	Source / Justification
NFR-TH-01	The system shall sustain the normal backend request workload.	Approximately 41.4 B requests/hour modeled average, within a publicly reported 39.6–43.2 B requests/hour operating range.	Derived from Spotify Engineering’s 11–12 M backend requests/second [16].
NFR-TH-02	The system shall support peak operating capacity without falling below the upper bound of the publicly reported continuous request range.	At least 43.2 B backend requests/hour during high-demand periods.	Capacity floor derived from 12 M requests/second [16]. The exact global production peak is not publicly disclosed.
NFR-TH-03	Campaign-related services shall remain operational throughout the initial high-demand Wrapped window.	3–4 hour peak window; campaign-service load on the order of tens of thousands of requests/second.	Spotify Engineering Wrapped load-testing report [17].
NFR-TH-04	The telemetry and personalization data path shall be capable of operating at platform-scale interaction volumes.	Platform-scale reference of approximately 3.4 T events and taste signals/day, equivalent to approximately 39.35 M/second on average.	Spotify Investor Day 2026 [22]. This is a platform-wide reference, not a music-only measurement.
NFR-TH-05	Catalog-ingestion processes shall support continuous large-scale music-release activity.	Global ecosystem reference greater than 100,000 newly released songs/day.	Spotify Newsroom [23]. Not every global release is necessarily ingested by Spotify.

Table 49: Throughput non-functional requirements

4.3 Response-Time Targets

Spotify does not publicly disclose complete end-to-end P95 and P99 SLOs for playback, search, recommendation, and advertising. Spotify Engineering does, however, explicitly use percentile-based latency measurements for production systems and playback performance [25, 26].

Therefore, the following values are proposed measurable requirements for the modeled system. They must

not be interpreted as publicly disclosed Spotify production SLOs.

ID	Operation	P95 Target	P99 Target	Requirement Rationale
NFR-RT-01	Audio playback initialization	< 500 ms	< 1 s	Playback initiation must remain sufficiently responsive to avoid a noticeable delay before media delivery begins.
NFR-RT-02	Music search query	< 300 ms	< 700 ms	Search is an interactive operation and results must be returned without interrupting catalog navigation.
NFR-RT-03	Recommendation request	< 500 ms	< 1 s	Personalized recommendation surfaces must be returned without introducing a noticeable user-interface delay.
NFR-RT-04	Advertising decision	< 300 ms	< 700 ms	Advertisement eligibility and selection must complete quickly enough to avoid unnecessary playback interruption.

Table 50: Proposed response-time non-functional requirements

4.4 Availability

Spotify does not publicly disclose availability SLAs for its consumer music-streaming services. Therefore, the availability targets defined in this section are requirements of the modeled system rather than claims about Spotify’s production SLAs.

As a public reference point, Spotify currently defines a monthly availability commitment of at least 99.5% for Spotify Portal for Backstage, a managed Spotify service [28]. This commitment is calculated from the proportion of successful valid requests and excludes planned and emergency maintenance.

For highly critical internal infrastructure, Spotify Engineering has also documented stricter service-level objectives. The Padlock key-management service was designed with a 99.95% availability SLO, with consideration of increasing this value to 99.99% [25].

These public values are used as reference points to define availability requirements according to the criticality of each service in the modeled music-streaming system.

ID	Service Domain	Availability Target	Maximum Downtime / Month	Required Behavior	Degraded Behavior
NFR-AV-01	Core music playback	99.99%	~4.38 min	Failures in secondary functionality shall not unnecessarily prevent already authorized music playback.	
NFR-AV-02	Authentication and account access	99.99%	~4.38 min	Existing valid sessions should remain usable whenever their current authorization can still be verified safely.	
NFR-AV-03	Search and catalog discovery	99.9%	~43.8 min	A partial failure shall not prevent already known playable content from being accessed through other supported navigation paths.	
NFR-AV-04	Personalization and recommendations	99.9%	~43.8 min	Failure of personalized recommendations shall not prevent basic catalog access or music playback.	
NFR-AV-05	Advertising services	99.9%	~43.8 min	Advertising-service degradation shall not cause failure of unrelated account or catalog operations.	
NFR-AV-06	Wrapped / annual-summary functionality	99.5%	~3.65 h	Failure of annual-summary functionality shall not prevent normal music streaming operations.	

Table 51: Availability non-functional requirements

The permitted downtime values use an average month of approximately 30.4 days. For example:

$$30.4 \times 24 \times 60 \times (1 - 0.9999) \approx 4.38$$

minutes of monthly downtime for a 99.99% availability target.

4.5 Data Volume and Growth

Spotify’s public information reports a catalog containing more than 100 million music tracks [21].

Spotify does not disclose an exact current storage footprint specifically for its music-only system. However, Spotify Engineering reports that petabytes of data are used for online workloads while exabytes reside in

its analytical data lake [18]. These values are platform-wide and are therefore used only as scale references for the modeled system.

Historical Spotify Engineering measurements also reported more than 350 TB of raw event data flowing through the event-delivery system every day [24]. That historical value corresponds to:

$$350 \text{ TB/day} \times 365 = 127,750 \text{ TB/year}$$

or approximately:

$$127.75 \text{ PB/year.}$$

Because that measurement dates from 2019 and Spotify’s event volume has grown substantially since then, it is treated as a conservative publicly documented lower-bound reference rather than a current exact annual-growth measurement.

ID	Requirement	Target / Benchmark	Source / Justification
NFR-DV-01	The music catalog shall support at least the current publicly reported Spotify catalog size.	More than 100 M tracks.	Spotify Investor Relations [21].
NFR-DV-02	The historical and analytical data layer shall be designed for at least exabyte-scale information.	At least exabyte-scale capacity.	Spotify Engineering reports exabytes in its current analytical data lake [18]. Platform-wide reference.
NFR-DV-03	Data required for low-latency online use cases shall be designed for petabyte-scale capacity.	Petabyte-scale online data.	Spotify Engineering reports petabytes supporting online use cases [18]. Platform-wide reference.
NFR-DV-04	The system shall plan for sustained annual data growth of at least the historically documented event-ingestion lower bound.	At least 127.75 PB/year raw-event-equivalent lower-bound reference.	Derived from Spotify’s historical 350 TB/day raw-event figure [24]. Current actual growth is expected to be higher and is not publicly disclosed in TB/year.

Table 52: Data-volume and growth non-functional requirements

4.6 Geographic Scope

Spotify currently reports availability in 184 markets [21]. The modeled system must therefore support users across different geographic regions, languages, licensing territories, network conditions, and local regulations.

Spotify does not currently publish an official Q2 2026 MAU distribution by country. For workload-distribution context only, a September 2026 secondary source based on Similarweb traffic estimates reports the United States at 24.71% of Spotify web/app traffic, Brazil at 13.64%, India at 4.23%, the United Kingdom at 3.92%, and Mexico at 3.23% [27].

These figures must not be interpreted as official Spotify MAU shares.

ID	Requirement	Target / Benchmark	Source / Justification
NFR-GEO-01	The system shall provide the modeled music-streaming functionality across the geographic scope publicly reported by Spotify.	184 markets.	Spotify Investor Relations [21].
NFR-GEO-02	Capacity planning shall account for geographically uneven traffic rather than assuming a uniform worldwide distribution.	September 2026 external traffic estimate: U.S. 24.71%, Brazil 13.64%, India 4.23%, U.K. 3.92%, Mexico 3.23%.	DemandSage using Similarweb traffic estimates [27]. This is not an official Spotify MAU distribution.
NFR-GEO-03	The response-time targets defined in Section 50 shall remain applicable to supported users regardless of the market from which the request originates, subject to the user's external network conditions.	184-market operation with the same service-level objectives.	Required to maintain consistent service quality across the stated geographic scope.

Table 53: Geographic-scope non-functional requirements

4.7 Non-Functional Requirements Summary

The publicly available evidence demonstrates that Spotify operates at a scale of hundreds of millions of active users, approximately 100 million concurrent clients, more than 100 million music tracks, tens of billions of backend requests per hour, trillions of daily platform interaction signals, and availability across 184 markets [19, 21, 16, 22].

For the modeled music-streaming system, the principal non-functional concerns are therefore:

- horizontal scalability for hundreds of millions of active users and approximately 100 million concurrent clients;
- sustained processing of tens of billions of backend requests per hour;
- high availability for core playback and account functionality;
- percentile-based response-time objectives for interactive operations;
- large-scale ingestion and processing of listening telemetry;
- storage capacity consistent with petabyte- and exabyte-scale data workloads;
- global operation across 184 markets;
- tolerance of concentrated high-demand periods such as Wrapped launches;

- continuous music-catalog ingestion and metadata processing; and graceful degradation so that failures in secondary functionality do not unnecessarily disable core music playback.

5 Workload Model

The workload model summarizes the quantitative operating profile used by this project. Values directly reported by Spotify are distinguished from derived calculations and proposed requirements.

Parameter	Value	Source / Justification
Expected active users	777 M MAUs	Official Spotify Q2 2026 figure [19]. Total registered-account count is not publicly disclosed.
Premium subscribers	300 M	Official Spotify Q2 2026 figure [19].
Ad-Supported MAUs	494 M	Official Q2 2026 filing [20].
Average concurrent clients	Approximately 100 M	Spotify Engineering states that around 100 million concurrent clients are served at any given moment [16].
Peak concurrent clients	Greater than or equal to 100 M capacity floor	An exact public peak-concurrency figure is not disclosed. The 100 M concurrent client measurement is therefore used as the minimum peak-capacity reference [16].
Average requests/hour	Approximately 41.4 B requests/hour	Derived from the midpoint of Spotify Engineering’s 11–12 M backend requests/second range: $11.5 \times 10^6 \times 3600$ [16].
Peak requests/hour	At least 43.2 B backend requests/hour	Capacity floor derived from the upper end of the publicly reported continuous range: $12 \times 10^6 \times 3600$ [16]. Exact platform-wide peak RPS is not publicly disclosed.
Peak duration	3–4 hours	Spotify Engineering reports that the first three to four hours of Wrapped launch produce some of the campaign’s most demanding traffic [17].
Campaign-specific peak workload	Tens of thousands of requests/second for the main Wrapped service	Spotify Engineering’s Wrapped load-testing target across the U.S., E.U., and Asia [17]. This is service-specific and is not treated as the platform-wide backend RPS figure.
Read/write ratio	Service-dependent; no verified platform-wide ratio is publicly disclosed	Spotify publishes examples of both strongly read-dominated services and write-heavy event/personalization workloads. For example, one critical read-dominated service reports fewer than 0.1% inserts or updates [25]. A single global numerical ratio would therefore be unsupported.

Parameter	Value	Source / Justification
Availability target	99.99% for core music playback	Proposed system requirement. Equivalent to approximately 4.38 minutes of maximum downtime per average month. Spotify Engineering has publicly documented availability objectives up to 99.99% for critical services [25].
Response-time target	Playback initialization P95 < 500 ms; P99 < 1 s	Proposed measurable requirement. Spotify Engineering publicly uses percentile-based latency measurements for playback and internal services [26, 25].
Music catalog volume	More than 100 M tracks	Spotify Investor Relations [21].
Analytical / historical data volume	Exabyte scale	Spotify Engineering reports exabytes in its analytical data lake [18]. This is a platform-wide scale reference because a music-only storage figure is not publicly disclosed.
Online data volume	Petabyte scale	Spotify Engineering reports petabytes supporting online use cases [18].
Annual data-growth reference	At least 127.75 PB/year historical raw-event lower-bound	Derived from Spotify’s historical 350 TB/day raw-event measurement: $350 \times 365 = 127,750$ TB/year [24]. Current exact TB/year growth is not publicly disclosed and is expected to be higher.
Daily platform interaction volume	Approximately 3.4 T events and taste signals/day	Spotify Investor Day 2026 [22]. Platform-wide reference; not claimed as a music-only event total.
Global music-release pressure	More than 100,000 new songs released/day	Spotify Newsroom [23]. This represents global music releases rather than confirmed Spotify uploads.
Geographic scope	184 markets	Spotify Investor Relations [21].
Geographic traffic distribution	U.S. 24.71%; Brazil 13.64%; India 4.23%; U.K. 3.92%; Mexico 3.23%	September 2026 Similarweb-based estimates reported by DemandSage [27]. These values represent traffic-share estimates rather than official Spotify MAUs.

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